MIL-P-52307A
7 April 1977
SUPERSEDING
MIL-P-52307
15 July 1963

MILITARY SPECIFICATION

PLOTTER, RECTANGULAR COORDINATES:

(COORDINATOGRAPH)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of precision plotters.

1.2 Classification. Plotters shall be of the following types, as specified (see 6.2):

Type I - Portable, with carrying case.

Type II - Table mounted (including light table).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-I-528	- Ink, Drawing, Waterproof, Black.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproof, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The plotter shall consist of a movable ordinate member, stationary abscissa member, ordinate carriage unit, and an abscissa carriage unit. The ordinate carriage unit, as a component of the ordinate member, shall have provisions for an insert holder and inserts.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3 Performance. The plotter drafting instrument shall draw pencil, pen, or scribe lines which shall not vary by more than 0.05 millimeter (mm) from a taut, thin wire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.4 Scale ratios. The following scale ratios shall be furnished:

1 mm = 1 mm
1 inch = 1 inch
1 : 6,000 metric
1 : 10,000 metric
1 : 12,500 metric
1 : 15,000 metric
1 : 17,000 metric
1 : 25,000 metric
1 : 50,000 metric

3.5 Identification marking. The plotter shall be identified in accordance with MIL-STD-130. The marking shall be applied to the plotter on plates conforming to MIL-P-514, Type I, Style 3, Composition C of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.6 Type I, portable, with carrying case. The Type I plotter shall be capable of being disassembled for transporting and shall weight not more than 90 pounds with carrying case and components. The instrument shall have a minimum working range of 650 mm in the ordinate "y" direction and 1000 mm in the abscissa "x" direction. The "y" axis shall be perpendicular to the "x" axis to within plus or minus 0.1 mm. The plotter shall be capable of plotting dimensionally related points to within plus or minus 0.1 mm.

3.6.1 Abscissa member. The abscissa member shall have sufficient weight to remain positioned on the drawing surface during operation of the ordinate member. The abscissa member shall have numbered scales engraved or permanently marked.

3.6.2 Abscissa carriage unit. The abscissa carriage unit shall be movable on the abscissa member and shall have a locking device, a magnifying lens with a tiltable mount for reading the abscissa vernier, and a vernier plate.

3.6.3 Ordinate member. The movable ordinate member shall have a large roller to support the free end and shall have numbered scales engraved or permanently marked. The ordinate member shall be mounted perpendicular to the abscissa member and shall be removable from the abscissa. The member, with its ordinate carriage unit, shall be freely movable over the entire design range of the plotter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.6.4 Ordinate carriage unit. The ordinate carriage unit shall be movable on the ordinate member and shall have a locking device, a magnifying lens with a tiltable mount for reading the ordinate vernier, an insert holder, and a vernier plate.

3.6.5 Insert holder. A holder (for inserts) shall be a part of the ordinate carriage unit and the following interchangeable inserts shall be furnished with the holder:

Pencil, lead (adjustable).
Pen, ruling (adjustable).
Point, pricking-scribing (adjustable).
Lens, magnifying (with a circle-positioning reticle).

3.6.6 Scribing points. Each plotter shall be furnished with rigid, round scribing points of solid carbide or corrosion-resisting steel with corundum or tungsten carbide tips, as specified (see 6.2). Scribing points furnished shall be in tip sizes 0.004 inch, 0.008 inch, and 0.010 inch.

3.6.7 Carrying case. A carrying case shall be furnished for the plotter and the tools and accessories.

3.6.8 Plate, orienting. A plate shall be furnished which will permit adjusting the instrument to an existing plan.

3.6.9 Vernier plates. The verniers shall subdivide the scales.

3.7 Type II, table mounted (including light table). The Type II plotter shall have a plotting range of 1200 mm in the ordinate direction and 1500 mm in the abscissa direction. The "y" axis shall be perpendicular to the "x" axis within plus or minus 0.05 mm. The plotter shall be capable of plotting dimensionally related points to within plus or minus 0.03 mm. Electrical components shall operate on 115-volt, single-phase, 60 Hertz. All electrical equipment and wiring shall conform to NFPA No. 70. A plate showing electrical data shall be securely affixed near the electrical input.

3.7.1 Abscissa member. The abscissa member shall have unnumbered engraved or permanently marked scales which are compatible with interchangeable numbered tapes and vernier dials.

3.7.2 Abscissa carriage unit. The carriage unit shall be movable on the abscissa member and shall have a locking device and a vernier dial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.7.3 Ordinate member. The movable ordinate member shall be mounted perpendicular to the abscissa member and shall have unnumbered engraved or permanently marked scales which are compatible with the interchangeable numbered tapes and vernier dials.

3.7.4 Ordinate carriage unit. The ordinate carriage unit shall be movable on the ordinate member and shall have a locking device, a vernier dial, and a holder.

3.7.5 Table. A table for the plotter shall be furnished. The table shall accommodate the plotting range of the instrument. The table shall have a glass top covered with a translucent plastic sheet and shall be illuminated from the back by fluorescent lights. The lights shall be replaceable. The table shall be mounted on a frame supported by legs that incorporate screws for leveling the plotter.

3.7.6 Holder. A holder (for inserts) shall be a part of the ordinate carriage unit and the following interchangeable inserts shall be furnished with the holder:

Pencil, lead (with adjustable chuck).
Pen, ruling (adjustable and carbide tipped).
Point, pricking-scribing (with adjustable pricking depth).
Microscope 7X magnification field view and 3.5 mm diameter, with a circle (positioning reticle) and pricker.
Scriber point holder (spring loaded).

3.7.7 Points. The following scribing points shall be furnished with the plotter:

Tool steel, single line.	- with cutting widths of 0.1 mm (.004 in.), 0.2 mm, and 0.3 mm. (Tolerance: plus or minus 0.0127 mm.)
Diamond- or corundum-tipped, single line.	- with cutting widths of 0.08 mm, 0.1 mm, 0.2 mm, and 0.3 mm. (Tolerance: plus or minus 0.0127 mm.)

3.7.8 Tapes. The tapes furnished with the plotter shall be used on the abscissa and ordinate members, shall have reels, shall be numbered for scale ratios specified in 3.6, shall be interchangeable, and shall have numbers advancing left to right.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

3.7.9 Dials. The dials used on the abscissa and ordinate carriages with scale ratios conforming to 3.4 shall have a vernier and shall have 2-way numerations. The millimeter dial shall have a least reading of 0.1 mm. The vernier used with the millimeter dial shall have a least reading of 0.01 mm.

3.7.10 Gage, orienting. A gage shall be furnished which will permit adjusting the instrument to an existing plan.

3.8 Workmanship. All parts, components, and accessories of the plotter, including castings, forgings, molded parts, stampings, bearings, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. All scale graduations engraved on the abscissa and ordinate members and verniers shall be legible. When a table is required, all surfaces shall be finished and preserved. The table frame and legs shall be rigidly constructed and joined to assure stability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

4.3 Quality conformance inspection.

4.3.1 Examination. Each plotter shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Each plotter shall be tested as specified in 4.4.2.1 through 4.4.2.5, as applicable. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The plotter shall be examined as specified herein for the following major defects:

101. Materials not as specified.
102. Scale ratios not as specified.
103. Abscissa member not as specified.
104. Abscissa carriage unit not as specified.
105. Ordinate member not as specified.
106. Ordinate carriage unit not as specified.
107. Insert holder missing or not as specified.
108. Scribing points missing or not as specified.
109. Carrying case not as specified.
110. Orienting plate not as specified.
111. Vernier plate not as specified.
112. Electrical components, equipment, or wiring not as specified.
113. Table not as specified.
114. Tapes not as specified.
115. Dials not as specified.
116. Orienting gage missing or not as specified.
117. Identification marking missing, incomplete, or illegible.
118. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Test conditions. Accuracy tests shall be performed in air-conditioned rooms maintained within plus or minus 2° F of any temperature in the 60° F to 75° F range. Relative humidity (rh) shall not change more than 5 percent during measurements. Measurements and comparison checks shall be made immediately after plotting and drafting to minimize dimensional changes in the plastic drafting materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

4.4.2.2 Drafting.

4.4.2.2.1 Pencil. Using a sharpened 3 H or 4 H pencil lead, draw gridlines 20 centimeters (cm) apart in the "x" direction across the entire length of a stable draft-coated polyester-base film having dimensions that approximate the plotters plotting range. Next draw gridlines in the "y" direction across the width of the drafting sheet. Check line straightness by using a 0.05- to 0.1-mm flexible wire weighted at each end with 1/4-pound (lb.) weights and stretched adjacent and parallel to the drawn line. Lines that are broken, ragged, or vary in distance from the stretched wire more than 0.05 mm shall constitute failure of this test.

4.4.2.2.2 Pen. Using the pen provided with the plotter and using ink conforming to TT-I-528, draw gridlines 20 cm apart in the "x" direction across the entire length of the stable draft-coated polyester-base film. Next, draw gridlines in the "y" direction across the entire width of the drafting sheet. Line straightness shall be checked as in 4.4.2.2.1. Lines that are broken, ragged, or vary in distance more than 0.05 mm from the stretched wire shall constitute failure of this test.

4.4.2.2.3 Scribing. Using the scribing point (0.1 mm or 0.004 inch) provided with the instrument, on stable draft-coated polyester-base film, scribe gridlines 20 cm apart in the "x" direction across the entire length of the scribe-coated sheet. Next, scribe gridlines in the "y" direction across the entire width of the sheet. Check line straightness as specified in 4.4.2.2.1. Lines that are broken, ragged, or vary in distance more than 0.05 mm from the stretched wire shall constitute failure of this test.

4.4.2.3 Angle (90 degrees - "y" to "x").

4.4.2.3.1 Type I plotter. To determine if the ordinate carriage ("y" axis) is positioned at 90 degrees to the abscissa carriage ("x" axis), tape a sheet of stable polyester-base film approximately 26 inches by 40 inches to the table top. Select a scale on the abscissa member and aline the zero of this scale with the corresponding zero of the abscissa vernier plate. Place the ordinate carriage unit at zero. This position should be such that the pricking point provided with the plotter is approximately 2 inches from the lower left corner of the taped draft sheet (see Figure 1, point "A"). Insert the pricking point into the holder located as a component of the ordinate carriage unit. Prick hole "A" into the draft sheet. After pricking the hole,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

raise the needle from the draft-sheet surface before moving the ordinate carriage unit. Loosen the locking device of the ordinate carriage unit and slide it slowly to the right until the pricking point is positioned approximately 2 inches from the right-hand edge of the draft sheet. Retighten the locking device. Prick hole "B". Leaving the ordinate carriage unit in this position, move the ordinate member slowly to the back edge of the draft sheet approximately 23 inches. Prick hole "C". Loosen the locking device. Leave the ordinate member in this position and move the ordinate carriage unit back to the starting zero and lock. Prick hole "D". Move the ordinate member from the test area. Using a magnifying glass and beam compass, adjust the compass-needle points until they exactly match the center-of-hole to center-of-hole distance on one diagonal of the rectangle ABCD. Move the beam compass to the other diagonal, place one compass point in a hole, and scribe a faint line near or through the related diagonal hole. A 40X shop microscope with measuring reticle shall be used to measure the distance from the scribed line to the center of the hole representing the end of the diagonal. A difference greater than plus or minus 0.1 mm shall constitute failure of this test.

4.4.2.3.2 Type II plotter. To determine if the ordinate member ("y" axis) is positioned at 90 degrees to the abscissa member ("x" axis), tape a sheet of stable polyester-base film approximately 42 by 42 inches to the table top. Select a scale printed on the numbered tape of the abscissa member and aline the zero mark of this scale with a graduation mark of the unnumbered scale on the abscissa member. This position should be such that the pricking point, provided with the plotter, is approximately 2 inches from the lower left corner of the taped draft sheet (see Figure 1, point "A"). Select the vernier scale dial having the same scale ratio as the tape, insert it in the abscissa member, and set the index mark to zero. Insert the pricking point into the holder located as a component of the ordinate carriage unit. Prick hole "A" into the draft sheet. After pricking the hole, raise the needle away from the draft-sheet surface before moving the ordinate carriage. Loosen the locking device of the ordinate carriage unit and slide it slowly to the right until the pricking point is positioned approximately 2 inches from the right-hand edge of the draft sheet. Retighten the locking device. Prick hole "B". Leaving the ordinate carriage unit in this position, move the ordinate member slowly to within 2 inches of the back edge of the draft sheet. Prick hole "C". Loosen the locking device. Leave the ordinate member in this position and move the ordinate carriage unit to the starting zero and lock. Prick hole "D". Move the ordinate member from the test area. Compare diagonal as specified in 4.4.2.3.1. A difference greater than plus or minus 0.05 mm shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

4.4.2.4 Plotting. The invar meter bar used for this test shall be certified by the National Bureau of Standards as being capable of marking measurements with a probable error not greater than plus or minus 0.005 mm.

4.4.2.4.1 Type I instrument.

4.4.2.4.1.1 "X" axis. Place the certified invar meter bar parallel to the abscissa member so that the positioning circle of the microscope reticle, provided with the plotter, is split by the graduation line of the invar meter bar at the zero mark. Using the 1 mm = 1 mm abscissa scale, set the related vernier plate to zero to aline with the scale zero. Move the ordinate member along the abscissa member to 10-cm on the invar scale. Split the microscope reticle circle with the 10-cm graduation line and record the reading of the vernier plate. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact abscissa measurement at that interval. Continue at 10-cm intervals to 100-cm. A difference greater than plus or minus 0.1 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test.

4.4.2.4.1.2 "Y" axis. Place the certified invar meter bar parallel to the ordinate member so that the positioning circle of the microscope reticle, provided with the plotter, is split by the graduation line of the certified invar meter bar at the zero mark. Using the 1 mm = 1 mm ordinate scale, set the ordinate vernier plate to zero to aline with the zero of the ordinate scale. Move the ordinate carriage unit along the ordinate member to the 10-cm mark on the invar scale. Split the microscope reticle circle with the invar meter bar graduation and record the reading of the vernier plate. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact ordinate measurement at that interval. Continue at 10-cm intervals to 60-cm. A difference greater than plus or minus 0.1 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:07Z
Check the source to verify that this is the current version before use.

4.4.2.4.2 Type II instrument.

4.4.2.4.2.1 "X" axis. Place the certified invar meter bar on the table top parallel to the abscissa member and positioned so that the circle of the microscope reticle, provided with the plotter, is split by the graduation line of the certified invar meter bar at the zero mark. Using the 1mm = 1 mm abscissa scale, set the abscissa tape to zero at the first graduation of the unnumbered scale on the "x" axis. Aline the vernier dial to zero. Move the ordinate member so as to split the microscope reticle with the certified invar meter bar 10-cm mark. Record the deviation, if any exists, of the vernier dial from 10-cm. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact abscissa measurement at that interval. Continue at 10-cm intervals to 100-cm. A difference greater than plus or minus 0.03 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test.

4.4.2.4.2.2 "Y" axis. Place the certified invar meter bar on the table parallel to the ordinate member so that the microscope reticle circle, provided with the plotter, is split by the zero graduation line of the meter bar. Using the same scale ratio as used for the "x" axis test, position the zero of the ordinate member tape to the first graduation on the unnumbered scale on the "y" axis. Aline the dial to zero. Move the ordinate carriage unit along the ordinate member to the 10-cm point on the invar scale. Record the deviation on the vernier dial from 10-cm. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact ordinate measurement at that interval. Continue at 10-cm intervals to 100-cm. A difference greater than plus or minus 0.03 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test.

4.4.2.5 Type II scribing point cutting width. The tool steel, diamond- and corundum-point widths specified shall be tested by scribing lines on a sheet of stable polyester-base film. With a 40X shop microscope (as specified in 4.4.2.3.1), measure the scribed line widths of the specified points. Deviations of more than plus or minus 0.0127 mm from the specified widths shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

119.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
120.	Components not cushioned and secured within the cases or boxes, as applicable, in a manner to prevent movement or damage for Level A.
121.	Table not unassembled as specified for Level A.
122.	Surfaces requiring a preservative not preserved and wrapped or covered as specified for Level A.
123.	Working surface of the table not protected as specified for Level A.
124.	Table not placed in box and contents not cushioned as specified for Level A.
125.	Box closure not as specified for Level A.
126.	More than one complete plotter packed together for Level A or Level B.
127.	Contents not secured within the shipping box in accordance with the referenced standard for Level A or Level B.
128.	Strapping not as specified for Level A.
129.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

5.1.1 Level A.

5.1.1.1 Type I. The components of the Type I plotter contained in the carrying case shall be preserved in accordance with MIL-P-116, Method IIb. The carrying case shall be considered the inner container, and the outer container shall be a close-fitting box conforming to PPP-B-636, W5c, style optional.

5.1.1.2 Type II. The components comprising a complete Type II plotter, except for the table (see 5.1.1.2.1), shall be contained in the contractor's cases or boxes, with the contents cushioned and secured to prevent movement or damage, and preserved in accordance with MIL-P-116, Method IIb. The contractor's case or box shall be considered the inner container, and the outer container shall be a close-fitting box conforming to PPP-B-636, W5c, style optional.

5.1.1.2.1 Table. The table shall be unassembled and components requiring the application of a contact preservative in accordance with MIL-P-116, shall be lightly coated with Type P-7 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Coated surfaces shall be covered or wrapped with barrier material conforming to MIL-B-121, Grade A, Type II, Class 2, secured in place with tape conforming to PPP-T-60, Type IV. The working surface of the table shall be covered with 1-inch-thick cushioning material conforming to PPP-C-843, Type II, Class A, and secured in place with tape specified herein. Each complete table shall then be placed in a close-fitting box conforming to PPP-B-636, V11c or V13c. The legs and any other loose components shall be wrapped with cushioning specified herein and shall be secured within the box in a manner to prevent movement or damage. Box closure shall be as specified for Method V in the appendix to the box specification.

5.1.2 Commercial. The components comprising a complete plotter shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. Each complete Type I or Type II plotter, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. The contents shall be secured within the box in a manner to prevent movement or damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A, except the box shall be Domestic Type. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. The components comprising a complete plotter shall be packed to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The plotter is used to prepare grid networks and plot dimensionally related points that are necessary for cartography.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of plotter required (see 1.2).
(c) Scribing point material required (see 3.6.6).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When other than Finish B strapping is required (see 5.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each plotter.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of plotters.

Custodians:
Army - ME
Air Force - 99

Review activity:
DSA - GS

Preparing activity:
Army - ME

Project 6675-0365

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

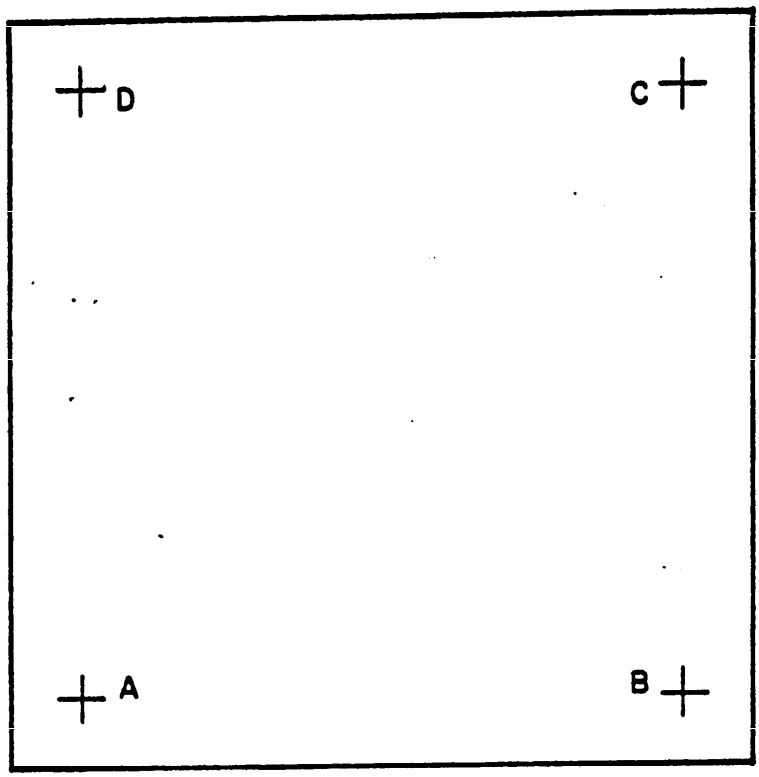


FIGURE 1. Dotting needle location(ABCD).

X-3224

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-P-52307A Plotter, Rectangular Coordinates: (Coordinatograph)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:07Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:07Z
Check the source to verify that this is the current version before use.